

readme_for_replication

readme.txt

Replication Data

Bart J. Bronnenberg, Jean-Pierre H. Dubé, and Matthew Gentzkow. The Evolution of Brand Preferences: Evidence from Consumer Migration.

Current Version: July 20, 2011.

HOW TO RUN

0. Delete the contents of each /output/ directory and its subdirectories but do not delete the subdirectories themselves
1. Obtain permission to use proprietary data from the contacts listed in "Contact Information for Proprietary Data" below.
2. Contact the authors with documentation of your authorization from the proprietary data providers. The authors will then provide you with the /external/ input files listed under "Omitted Input Files" below.
3. Place each of the input files in the appropriate ../external/ or ../input/ directory, as described in the OMITTED INPUT FILES section below.
4. Run "Brands (First Stage)/define_sample.do", "Brands (First Stage)/format_data.do", "Brands (First Stage)/format_data_bp.do", "Brands (First Stage)/RunFirstStage.m", and "Brands (First Stage)/RunFirstStage_bp.m" in sequence.
5. Run "Brands (IRI Elasticities)/preclean.do", "Brands (IRI Elasticities)/binaryLogit_IRI.do", "Brands (IRI Elasticities)/forecast_IRI.do", "Brands (IRI Elasticities)/cross_market.do", "Brands (IRI Elasticities)/tables.do", and "Brands (IRI Elasticities)/text.do" in sequence.
6. Run "Brands (CCA)/preclean.do" and "Brands (CCA)/analysis.do" in sequence.
7. Place all output files of /Brands (First Stage)/output/ into /Brands (Structural)/external/ and /Brands (Descriptive)/external/.
8. Run "Brands (Descriptive)/mu.m", "Brands (Descriptive)/preclean.do", "Brands (Descriptive)/preclean_mod_char.do", "Brands (Descriptive)/tables.do", "Brands (Descriptive)/figures.do", "Brands (Descriptive)/text.do", and "Brands (Descriptive)/online_appendix.do" in sequence.
9. Run "Brands (Structural)/ToMatlab.do", "Brands (Structural)/ShareChange.do", "Brands (Structural)/Text.do", "Brands (Structural)/RunStructuralModel.m", "Brands (Structural)/CounterFactuals.m", "Brands (Structural)/Tables.m", "Brands (Structural)/Figures.m", and "Brands (Structural)/TextTables.m" in sequence.

SYSTEM REQUIREMENTS

The above procedure has been tested on a 64-bit Windows 7 machine running Stata/MP 11 and Matlab 2010a.

FILE MANIFEST

/Brands (CCA)/: Code and output related to Consolidated Consumer Analysis
 /code/
 Contains code.
 analysis.do
 /output/

```

readme_for_replication
Stores output, including figures, text files, tables, and log files.
/figures/    histogram_cca.eps
              histogram_hs.eps
              predict_shares_[module].eps
              scatter.eps

analysis.log
preclean.log
rel_shares.csv
tables.txt

```

/Brands (Descriptive)/: Code and output related to descriptive analysis of brands.
/code/

```

Contains code.
figures.do
online_appendix.do
tables.do
text.do

```

/output/

```

Stores output, including figures, text files, tables, and log files.
/figures/    agemove_demo.eps
              agemove_none.eps
              agemove_recent_demo.eps
              agemove_recent_none.eps
              education_college_plus.eps
              income_over_median.eps
              male.eps
              secondary_is_migrant.eps
              summarystats.eps
              years_demo.eps
              years_none.eps
              years_none_unweighted.eps
              years0_none.eps
              years1_none.eps
              years25_demo.eps
              years25_none.eps

figures.log
mu.log
preclean.log
online_appendix.log
online_appendix_tables.txt
tables.log
tables.txt
text.log
texttables.txt

```

/Brands (First Stage)/: Code and output related to first stage of brands analysis.
/code/

```

Contains code.
define_sample.do
format_data.do
format_data_bp.do
input_param.txt
RunFirstStage.m
RunFirstStage_bp.m

```

/output/

```

Stores output, including figures, text files, tables, and log files.
NOTE: all data outputs of this directory are proprietary, and as such have
been omitted from the zipfile. Obtain proprietary data to view.
define_sample.log

```

/Brands (IRI Elasticities)/: Code and output to estimate the elasticity of substitution using the IRI data.
/code/

```

                                readme_for_replication
Contains code.
binaryLogit_IRI.do
cross_market.do
forecast_IRI.do
tables.do
text.do
/output/
Stores output, including figures, text files, tables, and log files.
coffee_xmkt.eps
elast_of_subs.txt
forecast.csv
forecast.dta
forecast_by_category.csv
forecast_panel.csv
forecast_sum.txt
linprob_price_effect.txt
regout.csv
regressions_xmkt.csv
tables.txt
texttables.txt

/Brands (Structural)/: Code and output related to structural analysis of brands.
/code/
Contains code.
CounterFactuals.m
Figures.m
FunctionMig.m
Get_AgeDist.m
Get_Beta_at.m
Get_Constants.m
Get_Response.m
RunStructuralModel.m
Tables.m
Text.do
Texttables.m
/output/
Stores output, including figures, text files, tables, and log files.
/figures/
    beta (a,t)bars.eps
    beta (a,t)checker.eps
    beta_{ij} - averagebars.eps
    beta_{ij} - averagechecker.eps
    dynamics_catchup.eps
    persistence.eps
    residbars.eps
    residchecker.eps

    tables.txt
    texttables.txt

```

```

=====
CONTACT INFORMATION FOR PROPRIETARY DATA
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NIELSEN HOMESCAN DATA

*** DATA ***

Data is from a custom online survey fielded by Nielsen PanelViews. The survey was fielded September 13, 2008 to October 1, 2008.

Original data file sent as UCLA Regional Bias Survey.zip from Nielsen/Grace Hyatt via e-mail on November 12, 2008.

Archive contained the following files:

280961_UCLA_final.tsv
280961_UCLA_Sept08.xls

readme_for_replication
UCLA Regional Purchase Bias Survey.doc

Original purchase data file sent as UCLA.zip from Nielsen/Todd Kaiser via e-mail (yousendit.com) on December 9, 2008. Revised file sent as 01.09.2009.zip on January 9, 2009.

Archive contained the following files:

UCLA_DATA_FILE.zip (which contained datafile UCLA_DATA_FILE)
UCLA_Trip_File.zip (which contained data file UCLA_TRIP_FILE)
NPD County Code.ppt
UCLA Data File Layout.xls
UCLA Family File Layout.xls
UCLA Trip File Layout
UCLA_Family_File.txt

*** CONTACT ***

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www.nielsen.com
<<http://www.nielsen.com/>>
<<http://www.nielsen.com/>>

TNS ADVERTISING

*** DATA ***

Brand_Top2_Dollars_2008_295519_export.csv:
Brand_Top2_Units_2008_295520_export.csv:

*** CONTACT ***

This data was downloaded from TNS Media Intelligence:
<http://www.tns-mi.com/>
1-800-497-8450

Using their data portal for AdSpender
(<http://adspender.tns-mi.com/ADSPENDER/Pages/Home.aspx?.pl=ADHome>).

AdSpender was purchased through Daniel Solomon:
Daniel Solomon
Service Bureau Representative
TNS Media Intelligence
100 Park Avenue 4th Floor
New York, NY 10017
Telephone: (212)-991-6055
Fax: (212)-991-6195
Hotline: 1-800-497-8450

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IRI

*** DATA ***

Data provided in CD form.

*** CONTACT ***

Data provided by IRI through the contact Mike Krueger.

<http://www.symphonyiri.com/>

1-800-497-8450

Mike Krueger

Information Resources Inc

150 N. Clinton, Chicago

Chicago, IL

TEL: 312.474.2524

FAX: 312.474.8865

MOB: 312.399.4597

OMITTED INPUT FILES

/Brands (CCA)/

EXTERNAL: ../external/by_brand.dta ; ../external/by_module.dta ;
../external/by_hh_brand.dta; ../external/demos.dta;
../external/CCA_Homescan_crosswalk.dta; ../external/brand_shares.dta

/Brands (Descriptive)/

EXTERNAL: ../external/survey.dta ; ../external/by_module.dta ;
../external/by_hh_brand.dta ; ../external/hh_module_month.dta ;
../external/sample_mod.csv; ../external/module_visibility.csv;
../external/module_top2brands_dollars_2008.dta
../external\b_demo_purch.csv; ../external\b_st_purch.csv;
../external\bnd_st_purch.csv; ../external\lhs_mig.csv; ../external\X_demo_mig.csv;
../external\X_stb_mig.csv; ../external\X_stc_mig.csv; ../external\b_demo_bp.csv;
../external\b_st_bp.csv; ../external\bnd_st_bp.csv; ../external\lhs_mig_bp.csv
INPUT: ../input/module_name_display.dta ; ../input/brand_name_display.dta

/Brands (First Stage)/

EXTERNAL: ../external/by_hh_pair.dta ; ../external\by_hh_module.dta ;
../external\by_module.dta ; ../external\survey.dta ; ../external\demos.dta
../external\module_top2same.csv

/Brands (IRI Elasticities)/

EXTERNAL: ../external/[product]_br_mkt_week.csv

/Brands (Structural)/

EXTERNAL: ../external\lhs_mig.csv ; ../external\X_demo_mig.csv ;
../external\X_stb_mig.csv ; ../external\X_stc_mig.csv ;
../external\lhs_mig_screen.csv ; ../external\lhs_mig_all.csv ;
../external\X_demo_mig_all.csv ;
../external\X_stb_mig_all.csv ; ../external\X_stc_mig_all.csv;
../external\b_st_purch.csv; ../external\b_demo_purch.csv;
../external\b_st_equiv.csv; ../external\b_demo_equiv.csv; ../external\b_st_exp.csv;
../external\b_demo_exp.csv; ../external\b_st_unit.csv;
../external\b_demo_unit.csv; ../external/survey.dta;
../external/module_visibility.csv; ../external/module_top2brands_dollars_2008.dta;
../external/module_chars.csv; ../external/hh_module_month.dta

DESCRIPTION OF OMITTED INPUTS

*/EXTERNAL/ FILES

readme_for_replication

by_brand.dta

- /Brands (CCA)/external/

Collapse purchase files from Nielsen Homescan proprietary data by brand (product group/module/brand).

by_module.dta

- /Brands (CCA)/external/; /Brands (Descriptive)/external/; /Brands (First Stage)/external/

Collapse purchase files from Nielsen Homescan proprietary data by product module.

by_hh_brand.dta

- /Brands (Descriptive)/external/; /Brands (CCA)/external/

Collapse purchase files from Nielsen Homescan proprietary data by household and brand.

hh_module_month.dta

- /Brands (Descriptive)/external/; /Brands (Structural)/external/

Collapse purchase files from Nielsen Homescan proprietary data by product module and month.

by_hh_module.dta

- /Brands (First Stage)/external/

Collapse purchase files from Nielsen Homescan proprietary data by household and product module.

by_hh_pair.dta

- /Brands (First Stage)/external/

Collapse purchase files from Nielsen Homescan proprietary data by household and brand pair.

survey.dta

- /Brands (Descriptive)/external/; /Brands (First Stage)/external/; /Brands (Structural)/external/

Dataset built from proprietary Nielsen Homescan data individual survey responses and state demographic information.

sample_mod.csv

- /Brands (Descriptive)/external/

List of product modules in the main sample used from the proprietary Nielsen Homescan data. Restricted to product modules in which we observe at least 5000 households making a purchase, and removing modules in which the top two brands are variants of the same brand.

demos.dta

- /Brands (First Stage)/external/; /Brands (CCA)/external/

Demographics data (e.g., age, education, employment, race) for households included in the Nielsen Homescan proprietary data.

module_top2same.csv

- /Brands (First Stage)/external/

Hand flagging of product modules from proprietary Nielsen Homescan data where top 2 brands are actually 2 varieties of the same brand.

lhs_mig.csv

- /Brands (Structural)/external/; /Brands (Descriptive)/external/

Purchase data for migrating households in the proprietary Nielsen Homescan data sample. Lists total purchases, equivalent units, quantity, and expenditure by household and produce module within the sample for the top two brands in each module. Excludes households for which the gap between the consumer's age and the sum of the number of years lived in birth state and the number of years lived in current state is greater than 5. Output of /Brands (First Stage)/.

readme_for_replication

X_demo_mig.csv

- /Brands (Structural)/external/; /Brands (Descriptive)/external/

Demographics for migrating households within the proprietary Nielsen Homescan data sample. Includes age, race, income, education, and employment. Excludes households for which the gap between the consumer's age and the sum of the number of years lived in birth state and the number of years lived in current state is greater than 5. Output of /Brands (First Stage)/.

X_stb_mig.csv

- /Brands (Structural)/external/; /Brands (Descriptive)/external/

Born-state for migrating households within the proprietary Nielsen Homescan data sample. Includes dummy for which state the head of the household was born in. Excludes households for which the gap between the consumer's age and the sum of the number of years lived in birth state and the number of years lived in current state is greater than 5. Output of /Brands (First Stage)/.

X_stc_mig.csv

- /Brands (Structural)/external/; /Brands (Descriptive)/external/

Current-state for migrating households within the proprietary Nielsen Homescan data sample. Includes dummy for which state the household currently lives in. Excludes households for which the gap between the consumer's age and the sum of the number of years lived in birth state and the number of years lived in current state is greater than 5. Output of /Brands (First Stage)/.

lhs_mig_screen.csv

- /Brands (Structural)/external/

Purchase data for households within the proprietary Nielsen Homescan data sample, keyed by household and product module. Flags state-modules with recorded sales for both top brands (based on both migrants and non-migrants). Includes share of top-brand purchases, and standard deviation of this share across modules and states. Output of /Brands (First Stage)/.

lhs_mig_all.csv

- /Brands (Structural)/external/

Purchase data for migrating households in the proprietary Nielsen Homescan data sample. Lists total purchases, equivalent units, quantity, and expenditure by household and produce module within the sample for the top two brands in each module. Output of /Brands (First Stage)/.

X_demo_mig_all.csv

- /Brands (Structural)/external/

Demographics for migrating households within the proprietary Nielsen Homescan data sample. Includes age, race, income, education, and employment. Output of /Brands (First Stage)/.

X_stb_mig_all.csv

- /Brands (Structural)/external/

Born-state for migrating households within the proprietary Nielsen Homescan data sample. Includes dummy for which state the head of the household was born in. Output of /Brands (First Stage)/.

X_stc_mig_all.csv

- /Brands (Structural)/external/

Current-state for migrating households within the proprietary Nielsen Homescan data sample. Includes dummy for which state the household currently lives in. Output of /Brands (First Stage)/.

b_st_purch.csv;

- /Brands (Structural)/external/; /Brands (Descriptive)/external/

State parameter estimates of the first-stage model, with purchases as quantity variable. Output of /Brands (First Stage)/.

b_demo_purch.csv;

readme_for_replication

- /Brands (Structural)/external/; /Brands (Descriptive)/external/
Demographic parameter estimates of the first-stage model, with purchases as quantity variable. Output of /Brands (First Stage)/.

b_st_equiv.csv

- /Brands (Structural)/external/
State parameter estimates of the first-stage model, with equivalent units as quantity variable. Output of /Brands (First Stage)/.

b_demo_equiv.csv

- /Brands (Structural)/external/
Demographic parameter estimates of the first-stage model, with equivalent units as quantity variable. Output of /Brands (First Stage)/.

b_st_exp.csv

- /Brands (Structural)/external/
State parameter estimates of the first-stage model, with expenditure as quantity variable. Output of /Brands (First Stage)/.

b_demo_exp.csv

- /Brands (Structural)/external/
Demographic parameter estimates of the first-stage model, with expenditure as quantity variable. Output of /Brands (First Stage)/.

b_st_unit.csv

- /Brands (Structural)/external/
State parameter estimates of the first-stage model, with units as quantity variable. Output of /Brands (First Stage)/.

b_demo_unit.csv

- /Brands (Structural)/external/
Demographic parameter estimates of the first-stage model, with units as quantity variable. Output of /Brands (First Stage)/.

module_top2brands_dollars_2008.dta

- /Brands (Structural)/external/
This file uses the Homescan-TNS crosswalk to merge on Homescan brand & module codes to TNS advertising expenditure data in terms of dollars, and then collapses expenditures to the module level.

module_visibility.csv

- /Brands (Structural)/external/
Hand coding of social visibility of modules by MG 7/4/10 based on joint decisions w/ BB and JPD; 2=high visibility (often consumed w/ others in social setting and consumed/served directly from branded packaging); 0=low visibility (personal care / medical / cleaning); 1=moderate visibility (everything else)

module_chars.csv

- /Brands (Structural)/external/
Answers provided about Homescan modules by RPs NP, YL, PD, and MT for questionnaire on product social availability.

[product]_br_mkt_week.csv

- /Brands (IRI Elasticities)/external/
IRI academic data: weekly, store-level data on brands for 31 [product] categories, including quantities sold for regular sales, feature advertisement, in-store display, and price reduction.

brand_shares.dta

- /Brands (CCA)/external/
A "long" shaped data set of brand shares from 1948.

CCA_Homescan_crosswalk.dta

readme_for_replication

- /Brands (CCA)/external/
Crosswalk from raw CCA names to common names for brands.

cca_hs_brand.dta

- /Brands (CCA)/temp/

Merge between proprietary Nielsen Homescan purchase data and non-proprietary Consolidated Consumer Analysis data containing purchase data keyed by brand and state. Also includes share and rank of an observation's purchases over state and module.

module_name_display.dta

- /Brands (Descriptive)/input/

Shortened display versions (20 characters max) of top-2 brand names from proprietary Nielsen Homescan data, limited to the 260 modules used in the analysis.

brand_name_display.dta

- /Brands (Descriptive)/input/

Shortened display versions (20 characters max) of module names from proprietary Nielsen Homescan data, limited to the 260 modules used in the analysis.

hh_mig.dta

- /Brands (Descriptive)/temp/

Migration data for migrating households in the proprietary Nielsen Homescan data sample. Includes current and born-in states/regions, current and migrating age, and gap between the consumer's age and the sum of the number of years lived in birth state and the number of years lived in current state.

hh_nonmig.dta

- /Brands (Descriptive)/temp/

Household data for non-migrating households in the proprietary Nielsen Homescan data sample. Includes current state/region and age.

modchar.dta

- /Brands (Descriptive)/temp/

Characteristics of product modules in proprietary Nielsen Homescan data, including dummies for highly visible and highly advertised modules (from proprietary TNS advertising data).

purch_mig.dta

- /Brands (Descriptive)/temp/

Purchase data for migrating households from proprietary Nielsen Homescan data, keyed by household and product module. Contains computed mu and beta estimates, as well as total purchases and purchase share for top brands.

purch_mig_bp.dta

- /Brands (Descriptive)/temp/

Purchase data for migrating households from proprietary Nielsen Homescan data, keyed by household and brand pair. Contains computed mu and beta estimates, as well as total purchases and purchase share for top brands.

purch_mig_month.dta

- /Brands (Descriptive)/temp/

Monthly purchase data for migrating households from proprietary Nielsen Homescan data, keyed by household, product module, and month. Contains computed mu and beta estimates, as well as total purchases and purchase share for top brands.

purch_nonmig.dta

- /Brands (Descriptive)/temp/

Purchase data for non-migrating households from proprietary Nielsen Homescan data, keyed by household and product module. Contains purchases, equivalent units, expenditures, and total units purchased of each of the top two brands.

stddev.dta

readme_for_replication

- /Brands (Descriptive)/temp/
Contains cross-state standard deviation of purchase shares, from non-migrating households within the proprietary Nielsen Homescan data.

pooldata.dta

- /Brands (IRI Elasticities)/temp/
Pooled data over all categories in the proprietary IRI data, collapsed to weekly frequency. Keyed by market, week, and category. Additional variables created for analysis include relative share and prices for the top two brands.

text.dta

- /Brands (Structural)/temp/
Annual typical share movement from the proprietary Nielsen Homescan data, including year-to-year absolute difference for purchase shares across modules with more than 5000 purchases. Keyed by product module.

hh_char.csv

- /Brands (Structural)/temp/
Survey age and number of years living in current household, from proprietary Nielsen Homescan survey data.

mod_char.csv

- /Brands (Structural)/temp/
Module characteristics from proprietary Nielsen Homescan data, including dummies for highly advertised, highly visible, highly purchased, and not for kids.

NOTES

1. Table Mapping:
For the tables, the map between the tags in files labeled tables.txt and the tables in the paper is as follows:

TABLE NAME	TABLE LOCATION
<tab:Migration>	Table 1
<tab:SumStat>	Table 2
<tab:Years>	Table 3
<tab:Years_bp>	Table 4
<tab:CCA>	Table 5
<tab:Struct>	Table 6
<tab:Struct_Catchup>	Table 7
<tab:Struct_Split>	Table 8
<tab:Robustness>	Appendix Table 1
<tab:ExtraSplits>	Appendix Table 2
<tab:mod260>	Appendix Table 3
<tab:YearsFGLS>	Online Appendix Table 1
<tab:Years_b_placebo>	Online Appendix Table 2
<tab:Generations>	Online Appendix Table 3

Files named texttables.txt provide replication of additional numbers cited in the paper.